

Sign and Conjugate Errors

Answer Key

Algebra 2

Quick Answers

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|---------------------------------------|---|
| 1. 61 | 5. the corrected quotient = $2 + 4i$, — |
| 2. $6 + 11i$ | 6. the corrected expansion = $21 - 20i$, — |
| 3. the corrected value = $3 + 8i$, — | 7. $11 - 10i$ |
| 4. $1 - 3i$ | 8. the corrected quotient = $11 + 2i$, — |
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What is verified

8 of 12 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSN-CN.A.2 – problems 1, 2, 3, 6, 7

HSN-CN.A.3 – problems 4, 5, 8

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

1. *If they got 11: treated i squared as $+1$, so the last term subtracted instead of adding*
6. *If they got 21: squared each term on its own and dropped the middle term*
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- 1.** Simplify $(6 - 5i)(6 + 5i)$.

Solution: The factors are conjugates, so the two middle terms $+30i$ and $-30i$ cancel and what is left is $36 - 25i^2$. Since $i^2 = -1$, that last term is $+25$, giving $36 + 25$. 61

- 2.** Simplify $(9 + 4i) - (3 - 7i)$.

Solution: The minus sign belongs to both parts of the second number: $9 + 4i - 3 + 7i$. Collecting, $(9 - 3) + (4 + 7)i$. $6 + 11i$

3. A student wrote $(8 + 2i) - (5 - 6i) = 3 - 4i$. (a) Correct it. (b) Name the error.

Solution (a): Distributing the minus sign to both parts gives $8 + 2i - 5 + 6i = (8 - 5) + (2 + 6)i$. $3 + 8i$

Part (b) — instructor-judged. The student distributed the minus sign to the real part only. The imaginary term of the second number is $-6i$, and subtracting it must give $+6i$, not $-6i$; the printed line kept $-6i$. Full credit requires naming that undistributed minus sign, not just writing the corrected value.

4. Simplify $\frac{10}{1 + 3i}$.

Solution: Multiply above and below by the conjugate $1 - 3i$. The denominator becomes $1 + 9 = 10$, so $\frac{10(1 - 3i)}{10} = 1 - 3i$. $1 - 3i$

5. A student divided $\frac{14 + 8i}{3 - 2i}$ and got $\frac{26 + 52i}{5}$. (a) Correct it. (b) Name the error.

Solution (a): The numerator is right: $(14 + 8i)(3 + 2i) = 42 + 28i + 24i + 16i^2 = 26 + 52i$. The denominator is where it goes wrong: $(3 - 2i)(3 + 2i) = 9 - 4i^2 = 9 + 4 = 13$. Dividing, $\frac{26 + 52i}{13}$. $2 + 4i$

Part (b) — instructor-judged. The conjugate product in the denominator is a *sum* of squares, because $-2i \cdot 2i = -4i^2 = +4$. The student wrote $9 - 4 = 5$ instead of $9 + 4 = 13$. Full credit names the denominator as the wrong value and gives 13.

6. A student expanded $(5 - 2i)^2 = 5^2 + (2i)^2 = 21$. (a) Correct it. (b) Name the error.

Solution (a): A square is a product of two binomials, so the middle term must appear:

$$\begin{aligned}(5 - 2i)^2 &= 25 - 10i - 10i + 4i^2 \\ &= 25 - 20i - 4\end{aligned}$$

$21 - 20i$

Part (b) — instructor-judged. The student squared each term separately and dropped the middle term $-20i$. Full credit names the missing middle term, or writes out all four products. Note that the student's real part happened to be right, which is exactly why a bare value check would not have caught this.

7. Simplify $(3 - 4i) - (-6 + i) + (2 - 5i)$.

Solution: Subtracting $-6 + i$ reverses *both* of its parts, so the expression becomes $3 - 4i + 6 - i + 2 - 5i$. Collecting, $(3 + 6 + 2) + (-4 - 1 - 5)i$. $11 - 10i$

8. A student's three-line solution to $\frac{20 + 15i}{2 + i}$ ends at $4 + 3i$. (a) Give the corrected quotient. (b) Which line is first wrong, and what rule does it break?

Solution (a): Both the numerator and the denominator gain the conjugate factor. Numerator: $(20 + 15i)(2 - i) = 40 - 20i + 30i - 15i^2 = 55 + 10i$. Denominator: $(2 + i)(2 - i) = 4 + 1 = 5$. Dividing, $\frac{55 + 10i}{5}$. $11 + 2i$

Part (b) — instructor-judged. Line 2 is the first wrong one. Line 1 is correct: multiplying by $\frac{2 - i}{2 - i}$ is multiplying by 1. But in line 2 the factor $2 - i$ was applied to the denominator only, and the numerator was carried down unchanged — which changes the value of the fraction. Full credit names line 2 and states that both parts of a fraction must gain the same factor. A student who checks by multiplying $4 + 3i$ back by $2 + i$ and getting $5 + 10i$ instead of $20 + 15i$ has produced the evidence and should be credited for it. Naming line 3 earns nothing: line 3 follows correctly from line 2.